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Studierichting: Informatica
Oefeningenreeks 5G nr. 6.8

$$f(x) = \sqrt{x} \text{ en } g(x) = x^2$$

$$x^2 = \sqrt{x} \Leftrightarrow x = x^4 \Leftrightarrow x = 0 \text{ of } x = 1$$

$$\begin{aligned} V &= \pi \int_0^1 (\sqrt{x^2} - x^4) dx = \pi \int_0^1 (x - x^4) dx \\ &= \pi \left[\left[\frac{x^2}{2} \right]_0^1 - \left[\frac{x^5}{5} \right]_0^1 \right] = \pi \left(\frac{1}{2} - \frac{1}{5} \right) = \frac{3\pi}{10} \end{aligned}$$

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References